Lecture 11

Cartan's calculus

Objectives. *Pullback compares a form at a moving point with a form at the starting point. Differentiating this comparison gives its Lie derivative. Cartan's formula expresses that derivative using exterior differentiation and insertion of a vector field. We prove the formula, including its time-dependent version, and apply it to volume preservation and Hamiltonian motion.*

By the end of this lecture you will be able to:

- Compute contraction with its first-slot sign convention.
- Prove Cartan's formula and differentiate a time-dependent pullback.
- Determine whether a local flow preserves a chosen volume form.
- Derive Liouville's theorem with the fixed Hamiltonian signs.

11.1 Contraction with a vector field

Let X be a smooth vector field on a smooth manifold M. For a k-form α with $k \geq 1$, its *interior product*, or *contraction*, with X fills the first argument:

$$(\iota_X \alpha)_p(v_1, \ldots, v_{k-1}) = \alpha_p(X_p, v_1, \ldots, v_{k-1}).$$

For a 1-form this is the function $\alpha(X)$. For a function f set $\iota_X f = 0$. The resulting form is smooth because its coordinate coefficients are sums of products of smooth coefficients of X and α.

Theorem 11.1 (Contraction rules)**.** *Contraction is real-linear, lowers degree by one, and satisfies*

$$\iota_X(\alpha \wedge \beta) = (\iota_X \alpha) \wedge \beta + (-1)^p \alpha \wedge \iota_X \beta, \qquad \iota_X^2 = 0,$$

where α has degree p. It is also linear over smooth functions in both X and the form.

Proof. For 1-forms $\alpha_1, \ldots, \alpha_k$, expand the determinant defining their wedge along its first column. This gives

$$\iota_X(\alpha_1 \wedge \cdots \wedge \alpha_k) = \sum_{j=1}^{k} (-1)^{j-1} \alpha_j(X) \alpha_1 \wedge \cdots \wedge \widehat{\alpha_j} \wedge \cdots \wedge \alpha_k.$$

The hat means that the factor is omitted. For the wedge of a block of p factors and a second block, split this sum according to the block containing the omitted factor. Omissions in the second block acquire $(-1)^p$. This is the stated product rule. Basis wedges span all forms, so bilinearity extends the identity to arbitrary forms. Applying contraction twice inserts X twice. Alternation makes this zero; on degrees zero and one it is zero by definition. Pointwise multilinearity proves both function-linear assertions and real-linearity. ∎

For example, $\iota_{\partial_x}(dx \wedge dy) = dy$. For the radial field $X = x\partial_x + y\partial_y$,

$$\iota_X(dx \wedge dy) = x\,dy - y\,dx.$$

Filling the last argument instead would reverse this sign.

11.2 Lie derivatives of forms and tensors

Let φ be the maximal local flow of X, supplied by Theorem 9.1. Near each $p \in M$, its maps are defined for all sufficiently small times. Pullback gives a form in the fixed exterior space at p, as in the figure.

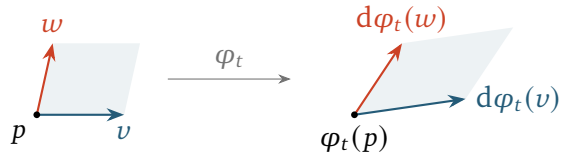


Figure 11.1: Pullback evaluates the form at $\varphi_t(p)$ on the transported vectors $d\varphi_t(v)$ and $d\varphi_t(w)$. The pullback at p gives this same evaluation.

The *Lie derivative of a form* is

$$(L_X\alpha)_p = \left.\frac{d}{dt}\right|_0 (\varphi_t^*\alpha)_p.$$

In coordinates the pullback has smooth coefficients jointly in time and position. Their time derivatives therefore define a smooth form. For a function, differentiating $f(\varphi_t(p))$ gives $L_X f = Xf$, agreeing with Lecture 9.

Theorem 11.2 (Lie derivative rules · *proved given smooth local flows*)**.** *The Lie derivative is real-linear and preserves degree. It satisfies*

$$L_X(\alpha \wedge \beta) = L_X\alpha \wedge \beta + \alpha \wedge L_X\beta, \qquad L_X d\alpha = d L_X\alpha.$$

Proof. Pullback preserves wedge by Theorem 10.3. Differentiating this identity at zero and using the ordinary product rule proves the first identity. No graded sign occurs because the derivative preserves degree. For the second, differentiate $\varphi_t^* d\alpha = d(\varphi_t^*\alpha)$. In a fixed chart, exterior differentiation differentiates coefficient functions in space. Time and space derivatives commute by joint smoothness. Thus the time derivative passes through d, proving the assertion. ∎

For example, the field $X = x\partial_x$ has flow $\varphi_t(x) = e^t x$. It gives $\varphi_t^* dx = e^t dx$, so $L_X dx = dx$.

Lie derivatives extend to tensors using the same comparison. For a diffeomorphism $f\colon M \to N$ and a tensor of type (r,s) on N, define f^*T by sending each tangent argument forward by df_p and each cotangent argument forward by the dual of $(df_p)^{-1}$. These arguments all lie at $f(p)$, where T is evaluated. Define $L_X T$ by differentiating $\varphi_t^* T$ at zero. This exists by the same smooth-coordinate argument. For vectors this pullback is $(\varphi_{-t})_* Y$, so it recovers $L_X Y = [X, Y]$ from Lecture 9. General tensor pullback here requires a diffeomorphism. Lecture 10's pullback formula for forms applies verbatim to covariant tensors along any smooth map.

11.3 Cartan's formula and time dependence

Theorem 11.3 (Cartan's formula · *proved given smooth local flows*)**.** *On differential forms,*

$$L_X = \mathrm{d}\iota_X + \iota_X \mathrm{d}.$$

Proof. Write $C_X = \mathrm{d}\iota_X + \iota_X\mathrm{d}$. The two graded product rules imply that C_X is an ordinary derivation. To see the cancellation explicitly, let $\deg\alpha = p$. In $C_X(\alpha\wedge\beta)$ the terms $(\iota_X\alpha)\wedge\mathrm{d}\beta$ have total coefficient $(-1)^{p-1} + (-1)^p = 0$. The terms $\mathrm{d}\alpha\wedge\iota_X\beta$ have coefficient $(-1)^p + (-1)^{p+1} = 0$. The remaining terms are $(C_X\alpha)\wedge\beta + \alpha\wedge C_X\beta$. The calculation also holds when $p = 0$, since contraction of a function is zero.

On functions, $C_X f = \iota_X \mathrm{d}f = Xf = L_X f$. On their differentials, $C_X\mathrm{d}f = \mathrm{d}(Xf) = L_X\mathrm{d}f$, using $\mathrm{d}^2 = 0$ and Theorem 11.2. Both sides depend only on the form near the point in question. For small time the trajectory stays in any chosen chart neighbourhood. Uniqueness makes the restricted field's flow agree with the original flow there. The Lie derivative therefore commutes with restriction to that chart. Locally every form is a sum of products of functions and coordinate differentials. The derivation rules therefore prove equality on every form in every chart. ∎

This is Tu, Theorem 20.10, with contraction in the first slot. For example, for $X = \partial_x$ and $\alpha = x\mathrm{d}y$, contraction gives zero while $\iota_X\mathrm{d}\alpha = \iota_X(\mathrm{d}x\wedge\mathrm{d}y) = \mathrm{d}y$. Thus $L_X\alpha = \mathrm{d}y$. Cartan's formula acts on forms. It does not replace the vector-field formula $L_XY = [X,Y]$.

For fixed α, the group law for an autonomous flow also gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\varphi_t^*\alpha = \varphi_t^* L_X\alpha.$$

Indeed, write $\varphi_{t+h}^* = \varphi_t^*\varphi_h^*$ and differentiate in h at zero. This identity holds wherever the indicated local maps exist.

A *time-dependent field* V_t has coefficients smooth jointly in $(t,p)\in I\times M$, where $I\subseteq\mathbb{R}$ is an open interval. Its evolution maps $\psi_{t,s}$ solve $\partial_t\psi_{t,s}(p) = (V_t)_{\psi_{t,s}(p)}$ with $\psi_{s,s}(p) = p$. Their local existence and joint smoothness follow by applying Theorem 9.1 to the field $(1, V_t)$ on $I\times M$. Uniqueness gives the local inverse $\psi_{s,t}$, so each evolution map is a diffeomorphism between its open domain and image. A *smooth family of forms* α_t has coefficients smooth in (t,p). Its derivative $\partial_t\alpha_t$ differentiates those coefficients in fixed, time-independent charts.

Proposition 11.4 (Time-dependent pullback · *proved given smooth local evolution maps*)**.** *Fix an initial time $s\in I$ and write $\psi_t = \psi_{t,s}$. On every region where these maps are defined,*

$$\frac{\mathrm{d}}{\mathrm{d}t}\psi_t^*\alpha_t = \psi_t^*\big(L_{V_t}\alpha_t + \partial_t\alpha_t\big) = \psi_t^*\big(\mathrm{d}\iota_{V_t}\alpha_t + \iota_{V_t}\mathrm{d}\alpha_t + \partial_t\alpha_t\big).$$

Proof. Fix t and define $\eta_h = \psi_{t+h}\circ\psi_t^{-1}$ on the current image, shrinking neighbourhoods and the range of h if necessary. Its initial value is the identity and $\partial_h\eta_h|_0 = V_t$. Joint smoothness gives the coordinate expansions

$$\eta_h(q) = q + hV_t(q) + O(h^2), \qquad D\eta_h(q) = I + hDV_t|_q + O(h^2).$$

These agree to first order with the autonomous flow of the fixed field V_t. The coordinate pullback formula uses just these first derivatives in h, so $\partial_h(\eta_h^*\alpha_t)|_0 = L_{V_t}\alpha_t$. Now $\psi_{t+h}^*\alpha_{t+h} = \psi_t^*\eta_h^*\alpha_{t+h}$. The family $\eta_h^*\alpha_{t+k}$ is jointly smooth in h and k. The chain rule along $h = k$ adds its two partial derivatives at zero: $L_{V_t}\alpha_t$ and $\partial_t\alpha_t$. The fixed pullback ψ_t^* carries their sum back

to the starting point. Cartan's formula gives the second equality. No autonomous group law for ψ_t was used. ∎

For example, $\psi_t(x,y) = (x + t^2 y, y)$ is generated from time zero by $V_t = 2ty\partial_x$. For $\alpha_t = tx\mathrm{d}y$ its pullback is $t(x + t^2 y)\mathrm{d}y$. The derivative is $(x + 3t^2 y)\mathrm{d}y$. Here $L_{V_t}\alpha_t = 2t^2 y\mathrm{d}y$ and $\partial_t \alpha_t = x\mathrm{d}y$. Pulling their sum back gives precisely that same derivative. Omitting the explicit time derivative would give a different answer.

11.4 Volume forms and divergence

On an m-manifold, a *volume form* μ is a smooth m-form that is nonzero at every point. Since the top exterior space is one-dimensional, there is a unique smooth function $\mathrm{div}_\mu X$ satisfying

$$L_X \mu = (\mathrm{div}_\mu X)\mu.$$

This function is the *divergence relative to* μ. Smoothness follows by dividing the local coefficient of $L_X\mu$ by the nowhere-zero coefficient of μ. A flow *preserves* a form if its pullback equals that form on every time map's domain. Such a form is called *invariant* under the flow.

Proposition 11.5 (Divergence and volume preservation · *proved given smooth local flows*). *In local coordinates, for $\mu = \rho\mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^m$, with $\rho \neq 0$,*

$$\mathrm{div}_\mu X = \frac{1}{\rho}\sum_{i=1}^{m} \partial_i(\rho X^i).$$

The local flow of X preserves μ if and only if $\mathrm{div}_\mu X = 0$ everywhere.

Proof. Since there are no forms of degree $m+1$, Cartan gives $L_X\mu = \mathrm{d}\iota_X\mu$. The contraction formula gives

$$\iota_X\mu = \sum_i (-1)^{i-1}\rho X^i \mathrm{d}x^1 \wedge \cdots \wedge \widehat{\mathrm{d}x^i} \wedge \cdots \wedge \mathrm{d}x^m.$$

After differentiation, only the x^i derivative of the ith coefficient survives. Moving $\mathrm{d}x^i$ to its ordered place gives another $(-1)^{i-1}$. Thus the resulting top-form coefficient is $\sum_i \partial_i(\rho X^i)$. Dividing by ρ proves the formula. If divergence vanishes, the finite-time pullback derivative is zero. Along each allowed time interval it is constant and equals μ at time zero. Conversely, differentiating invariance at zero gives $L_X\mu = 0$, hence zero divergence. ∎

For Euclidean volume, $\rho = 1$ gives the usual sum $\sum_i \partial_i X^i$. For example, $X = x\partial_x + y\partial_y$ has divergence 2 relative to $\mathrm{d}x \wedge \mathrm{d}y$. Its flow multiplies that form by e^{2t}. On the punctured plane choose instead $\mu' = (x^2+y^2)^{-1}\mathrm{d}x \wedge \mathrm{d}y$. The same field gives

$$\mathrm{div}_{\mu'} X = (x^2+y^2)\left(\partial_x \frac{x}{x^2+y^2} + \partial_y \frac{y}{x^2+y^2}\right) = 0.$$

The two derivatives sum to $2/r^2 - 2r^2/r^4 = 0$, where $r^2 = x^2 + y^2$. Thus its flow preserves μ' while expanding Euclidean area. The choice of volume form matters.

11.5 Symplectic forms and Liouville's theorem

On a $2n$-manifold with $n \geq 1$, a *symplectic form* ω is a closed smooth 2-form that is *nondegenerate*: if $\omega_p(v, w) = 0$ for every $w \in T_pM$, then $v = 0$. For a smooth Hamiltonian function $H\colon M \to \mathbb{R}$, its *Hamiltonian field* X_H is determined by

$$\iota_{X_H}\omega = \mathrm{d}H.$$

Nondegeneracy makes $v \mapsto \omega_p(v, \cdot)$ an isomorphism. Its inverse has smooth matrix entries in charts, so the field exists uniquely and is smooth. The equation $\iota_{X_H}\omega = \mathrm{d}H$ is the sign convention of *Theoretical Mechanics*, notation §1, recalled here unchanged.

Corollary 11.6 (Liouville · *proved given smooth local flows*)**.** *For a symplectic form and its Hamiltonian field,*

$$L_{X_H}\omega = 0, \qquad L_{X_H}(\omega^n) = 0.$$

Consequently its local flow preserves both the symplectic form and the volume form $\omega^n/n!$.

Proof. Cartan's formula and the defining Hamiltonian equation give $L_{X_H}\omega = \mathrm{d}^2H + \iota_{X_H}\mathrm{d}\omega$. Both terms vanish because $\mathrm{d}^2 = 0$ and ω is closed. The derivation rule then gives zero on the n-fold wedge power ω^n. The finite-time pullback identity proves preservation.

To verify that the top wedge is nonzero, choose tangent vectors a, b with $\omega(a, b) = 1$. Every tangent vector v splits into a component $\omega(v, b)a - \omega(v, a)b$ in their plane and a component orthogonal to that plane for ω, meaning that its pairing with both a and b is zero. The restriction to the complementary subspace is nondegenerate, since a vector orthogonal to it and to the plane is orthogonal to the whole space. Repeat in the complement. In the resulting dual basis α^j, β^j, $\omega = \sum_j \alpha^j \wedge \beta^j$. Its nth power is $n!\alpha^1 \wedge \beta^1 \wedge \cdots \wedge \alpha^n \wedge \beta^n$, which is nonzero. Thus the preserved top form is indeed a volume form. ∎

11.6 Hamiltonian motion and an incompressible fluid

On $T^*\mathbb{R}^n$, write the position coordinates as q^i and the cotangent coefficients as p_i. Here θ denotes the canonical 1-form. The symplectic form is ω. Their coordinate expressions are

$$\theta = p_i \mathrm{d}q^i, \qquad \omega = \sum_i \mathrm{d}q^i \wedge \mathrm{d}p_i.$$

The positions q^i translate the lower coordinate labels of *Theoretical Mechanics*. The momenta retain their covector indices.

Example 11.7 (The canonical signs)**.** Differentiating gives $\mathrm{d}\theta = \sum_i \mathrm{d}p_i \wedge \mathrm{d}q^i = -\omega$. For $X = \dot{q}^i\partial_{q^i} + \dot{p}_i\partial_{p_i}$, contraction gives

$$\iota_X\omega = \sum_i (\dot{q}^i \mathrm{d}p_i - \dot{p}_i \mathrm{d}q^i).$$

Equating this with $\mathrm{d}H$ gives Hamilton's equations $\dot{q}^i = \partial H/\partial p_i$ and $\dot{p}_i = -\partial H/\partial q^i$. The form is closed because its coefficients are constant, and the displayed contraction shows nondegeneracy. Liouville therefore proves preservation of this form and of its top wedge volume.

Recall the canonical *Poisson bracket* of smooth functions:

$$\{f, g\} = \sum_i \left(\frac{\partial f}{\partial q^i}\frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i}\frac{\partial g}{\partial q^i} \right).$$

The displayed Hamiltonian field acts on a smooth function h by $X_H h = \{h, H\}$, as its coordinate coefficients show. For one degree of freedom, take $f = q^2/2$ and $g = p^2/2$. Their fields are $X_f = -q\partial_p$ and $X_g = p\partial_q$. The coordinate bracket is $[X_f, X_g] = -q\partial_q + p\partial_p$. The fixed Poisson bracket gives $\{f, g\} = qp$, whose field is $X_{qp} = q\partial_q - p\partial_p$. Thus $[X_f, X_g] = -X_{\{f,g\}}$, computed using Lecture 9's bracket $[X, Y]h = X(Yh) - Y(Xh)$.

The harmonic oscillator with $H = (q^2 + p^2)/2$ has field $p\partial_q - q\partial_p$. Its complete flow is

$$\varphi_t(q, p) = (q\cos t + p\sin t, -q\sin t + p\cos t).$$

Differentiation gives $(\dot{q}, \dot{p}) = (p(t), -q(t))$ and the initial value is (q, p). From $(1, 0)$ the flow reaches $(0, -1)$ at time $\pi/2$. The phase-plane motion is clockwise, as shown in the figure; the origin is fixed.

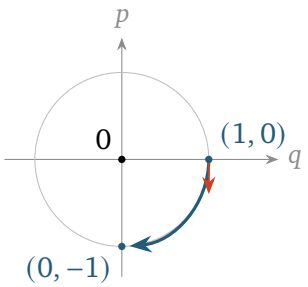


Figure 11.2: Positive-time harmonic-oscillator motion is clockwise in the (q, p) plane. The orbit through $(1, 0)$ reaches $(0, -1)$ after a quarter period; the origin is stationary.

For a fluid in Euclidean space, let v_t be its smooth velocity field. Constant-density *incompressibility* is the condition $\sum_i \partial_i v_t^i = 0$ at every point and time. With Euclidean volume $\mu = \mathrm{d}x^1 \wedge \cdots \wedge \mathrm{d}x^m$, this is exactly $L_{v_t}\mu = 0$. The time-dependent pullback formula, with $\partial_t \mu = 0$, therefore proves $\psi_t^* \mu = \mu$ wherever fluid trajectories exist. The moving fluid preserves this chosen local volume element.

Self-check. For a smooth function $f : \mathbb{R} \to \mathbb{R}$ and $X = x\partial_x$, compute $L_X(f\mathrm{d}x)$.

Answer. Cartan gives $\mathrm{d}\iota_X(f\mathrm{d}x) = \mathrm{d}(xf) = (xf' + f)\mathrm{d}x$. The other term is zero because every 2-form on $\mathbb{R}$ vanishes.

Summary. Contraction inserts a vector in the first slot and is a graded antiderivation. The Lie derivative of forms differentiates flow pullback. Cartan expresses it as $\mathrm{d}\iota_X + \iota_X\mathrm{d}$. A time-dependent form contributes its explicit time derivative as well. Zero divergence relative to a chosen volume form is equivalent to local flow preservation of that form. A Hamiltonian flow preserves its symplectic form and the associated top wedge volume. Lecture 12 integrates forms and relates their exterior derivatives to boundaries.

Reading. Tu, *An Introduction to Manifolds*, §20.3–§20.5, especially Theorem 20.10, pp. 229–230.